Delin An

Scientific AI | Generative Modeling | 3D Computer Vision | Medical Imaging

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RESEARCH PROFILE

Postdoctoral researcher developing data-efficient generative and representation-learning methods that unify images, masks, and geometry. My work spans controllable 3D medical-image synthesis, computer vision, and embodied perception, with an emphasis on reliable systems that connect AI to scientific, clinical, and physical applications.

ACADEMIC APPOINTMENT

Cornell University — 2026 - Present

Postdoctoral Researcher | Ithaca, NY

- Develop **scientific machine-learning and generative-modeling methods** for geometry-rich problems in biomedicine and computational science.
- Advance multimodal 3D representation learning for **controllable physical modeling**, uncertainty-aware prediction, and simulation-ready geometry.

EDUCATION

University of Notre Dame — Sep 2022 - May 2026

Ph.D., Computer Science and Engineering | GPA: 3.8/4.0 | Generative AI, 3D vision, medical imaging

Xi'an Jiaotong University — Sep 2019 - Jun 2022

M.Eng., Mechanical Engineering | GPA: 86.28/100 | Robotics, perception, embedded control

University of Electronic Science and Technology of China — Sep 2015 - Jun 2019

B.Eng., Mechanical Engineering | GPA: 3.61/4.0 | Top 10%

SELECTED RESEARCH CONTRIBUTIONS

Sketch2CT - Multimodal 3D Medical Volume Generation — CVPR 2026

Generative AI | Multimodal learning | Medical imaging

- Created a **two-stage latent-diffusion framework** that converts a user sketch plus text into anatomically coherent 3D masks and medical volumes, using capsule attention, text-guided sketch refinement, and cross-modal feature fusion.
- Achieved the strongest structural fidelity on **four public CT/MRI datasets**; synthetic-data training reached **Dice scores up to 0.904** and closely approached real-data training on three datasets.

AortaDiff and Automated Patient-Specific Modeling — VIS / Sci. Adv. 2025

3D geometry | Diffusion | Computational hemodynamics

- Designed **end-to-end pipelines** from volumetric scans to centerlines, contours, smooth multi-branch surfaces, and **CFD-ready meshes**; integrated Bayesian segmentation, diffusion, GNN-LDDMM deformation, NURBS, and geometry optimization.
- Generated **high-fidelity patient-specific models** across healthy and pathological aortas, including aneurysm and coarctation, under limited-data settings.

Sli2Vol+ - Label-Efficient 3D Segmentation — WACV 2025

Correspondence learning | Weak supervision

- Developed object-estimation-guided correspondence flow and pseudo-label refinement for single-slice-to-volume annotation; achieved **78.9 mean Dice** across CT/MRI benchmarks, **4.0 points** above the strongest reported baseline.

SurfPatch - Exploratory Surface Analytics — TVCG 2025

Geometric matching

- Introduced **multiscale vertex-to-patch-to-surface matching** with an **interactive UMAP interface** for partial queries across complex flow surfaces.

PEER-REVIEWED PUBLICATIONS

1. "Sketch2CT: Multimodal Diffusion for Structure-Aware 3D Medical Volume Generation." *CVPR*, pp. 37600-37610, 2026. **First author.**
2. "AortaDiff: Volume-Guided Conditional Diffusion Models for Multi-Branch Aortic Surface Generation." *IEEE TVCG*, 32(1):922-932, 2026; IEEE VIS 2025. **First author.**
3. "AI-Powered Automated Model Construction for Patient-Specific CFD Simulations of Aortic Flows." *Science Advances*, 11(36):eadw2825, 2025. **Co-first author.**
4. "SurfPatch: Enabling Patch Matching for Exploratory Stream Surface Visualization." *IEEE TVCG*, 31(6):3771-3782, 2025. **First author.**
5. "Sli2Vol+: Segmenting 3D Medical Images Based on an Object Estimation Guided Correspondence Flow Network." *WACV*, pp. 3624-3634, 2025. **First author.**
6. "Hierarchical LoG Bayesian Neural Network for Enhanced Aorta Segmentation." *ISBI*, pp. 1-5, 2025. **First author; oral presentation.**
7. "Environmental Obstacle Detection and Localization Model for Cable-Driven Exoskeleton." *IEEE UR*, pp. 64-69, 2022. **First author.**
8. "Research on Lower Limb Rehabilitation Exoskeleton Control Based on Improved Dynamic Motion Primitives." *IEEE ICARM*, pp. 443-448, 2022. **Co-author.**
9. "The Design and Implementation of Human Motion Capture System Based on CAN Bus." *IEEE UR*, pp. 415-420, 2020. **Co-author.**

ROBOTICS AND EMBODIED SYSTEMS

Exoskeleton Terrain Perception and Adaptive Gait Control 2020 - 2022

Computer vision | Perception-to-control integration

- Built a **skip-connected fully convolutional network** for pixel-level terrain recognition and boundary perception, then connected its outputs to **finite-state gait switching**.
- Integrated perception and adaptive control to automate gait-pattern changes across outdoor terrain and reduce wearer effort.

Cable-Driven Ankle Exoskeleton 2020 - 2021

Wearable robotics | Mechanical design | Embedded control

- Designed, FEA-validated, fabricated, and assembled a **lightweight exoskeleton prototype**; implemented impedance control on an **STM32 platform**.

AWARDS AND HONORS

- Scientific Artificial Intelligence (SAI) Graduate Fellowship Pilot, University of Notre Dame, 2025 - inaugural cohort and first/only recipient from Computer Science and Engineering.
- Graduate School Professional Development Award (Zahm Fund), University of Notre Dame, 2025.
- First-Class Scholarship (2020) and Special Scholarship (2019), Xi'an Jiaotong University.
- Grand Prize, Robot Creativity Contest; Third Prize, National Graduate Robotics Contest.
- Outstanding Graduate (2019); People's Scholarships (2016, 2017), UESTC.

TECHNICAL STRENGTHS

Generative AI and representation learning: 3D latent diffusion, controllable generation, transformers, multimodal fusion, distributed and mixed-precision training, Bayesian inference.

Computer vision and medical AI: 3D segmentation, correspondence learning, weak supervision, uncertainty quantification, CT/MRI synthesis, label-efficient learning.

Geometry and scientific computing: surface reconstruction, NURBS, GNN-LDDMM, geometric matching, PyVista, ParaView, OpenFOAM, CFD-ready meshing.

Robotics and engineering: CNN terrain perception, finite-state gait control, impedance control, STM32/Keil, SolidWorks, ANSYS, MATLAB, CAD/FEA, physical prototyping.

Programming and platforms: Python, PyTorch, MONAI, NumPy, SciPy, C/C++, Git, Linux.